INTRODUCTION

In many contexts in investments, we want to assess the strength of the linear relationship between two variables—that is, we want to evaluate the **correlation** between them. A significance test of a correlation coefficient allows us to assess whether the relationship between two random variables is the result of chance. Lesson 1 covers a parametric and a non-parametric approach to testing the correlation between two variables. If we decide that the relationship does not result from chance, then we can use this information in modeling or forecasting using regression models or machine learning covered in later Learning Modules.

When faced with categorical or discrete data, however, we cannot use the methods discussed in the first lesson to test whether the classifications of such data are independent. If we want to test whether there is a relationship between categorical or discreet data, we can perform a test of independence using a nonparametric test statistic. The second lesson covers the use of contingency tables in implementing this non-parametric test.

Learning Module Overview

- There are three ways to formulate hypotheses. Let θ indicate the population parameters:

  1. Two-sided alternative: H_0: $\theta = \theta_0$ versus Ha: $\theta \neq \theta_0$
  2. One-sided alternative (right side): H_0: $\theta \leq \theta_0$ versus Ha: $\theta > \theta_0$
  3. One-sided alternative (left side): H_0: $\theta \geq \theta_0$ versus Ha: $\theta < \theta_0$

  where θ_0 is a hypothesized value of the population parameter and θ is the true value of the population parameter.
- A parametric test is a hypothesis test concerning a population parameter or a hypothesis test based on specific distributional assumptions. In contrast, a nonparametric test either is not concerned with a parameter or makes minimal assumptions about the population from which the sample comes.
- A nonparametric test is primarily used when data do not meet distributional assumptions, when there are outliers, when data are given in ranks, or when the hypothesis we are addressing does not concern a parameter.
- In tests concerning correlation, we use a t-statistic to test whether a population correlation coefficient is different from zero. If we have n observations for two variables, this test statistic has a *t*-distribution with $n - 2$ degrees of freedom.
- The Spearman rank correlation coefficient is calculated on the ranks of two variables within their respective samples.
- A chi-square distributed test statistic is used to test for independence of two categorical variables. This nonparametric test compares actual frequencies with those expected on the basis of independence. This test statistic has degrees of freedom of $(r - 1)(c - 1)$, where r is the number of categories for the first variable and c is the number of categories of the second variable.

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